Inverted Maximum Entropy Classification on Blocking Candidate Sets using `automatedRecLin`

Methodological Specification for Binary and Continuous Parametric MEC

Methodological draft for StatsClaw/Codex implementation

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1 Purpose and Scope

This document specifies a methodological extension of the unsupervised Maximum Entropy Classification (MEC) procedure for record linkage so that it can be applied directly to a candidate-pair space produced by the `blocking` package. The goal is to avoid a separate two-stage workflow in which an unsupervised MEC model is first trained on one pair space and then used as a prediction model on another pair space. Instead, the proposed procedure trains and applies MEC on the same blocking-based candidate-pair space.

The method is intended for two-file record linkage between datasets A and B. It can be adapted later to other settings, but the present specification assumes that a valid linkage set must satisfy the usual one-to-one constraint: each record from A and each record from B can appear in at most one selected matched pair.

Only two comparison-model families are considered here:

(i) the binary MEC model based on binary agreement indicators;

(ii) the continuous parametric MEC model based on hurdle Gamma comparison densities.

The specification deliberately focuses on the statistical algorithm, required methodological changes, outputs, and validation invariants. It does not prescribe repository layout, low-level data structures, or programming-language-specific implementation details.

2 Classical MEC Baseline

Let A and B denote two files without duplicates within each file. In the classical MEC formulation, the full comparison space is

$$\Omega = A \times B,$$

which is partitioned into the set of true matches M and the set of true nonmatches U:

$$\Omega = M \cup U.$$

For each pair $(a, b) \in \Omega$, let

$$\gamma_{ab} = (\gamma_{ab}^1, \ldots, \gamma_{ab}^K)'$$

be the comparison vector based on K key variables. Let

$$m(\gamma) = f(\gamma \mid (a, b) \in M), \qquad u(\gamma) = f(\gamma \mid (a, b) \in U)$$

denote the match and nonmatch comparison distributions. Classical MEC uses the density ratio

$$r(\gamma) = \frac{m(\gamma)}{u(\gamma)}.$$

Large values of $r(\gamma)$ indicate that a pair is more likely to be a match.

A MEC classification set is a subset of record pairs satisfying the one-to-one constraint. For a requested size n^*, the classical MEC set is obtained by sorting pairs in descending order of $r(\gamma_{ab})$ and greedily adding a pair when neither of its records has already been selected. The procedure stops once n^* pairs have been selected.

In the unsupervised classical algorithm, the full pair space Ω is treated as highly imbalanced and dominated by nonmatches. This motivates estimating u from all pairs in Ω, estimating m iteratively from the current matched set, estimating the number of matches, and constructing a new MEC set in descending order of $r = m/u$.

3 Blocking Candidate Space

After blocking, MEC should not be applied to the full Cartesian product unless the workflow explicitly constructs such a product. Instead, define

$$\Omega_{\mathcal{B}}$$

as the candidate-pair space passed from the blocking stage to the linkage stage. Its size is

$$N = |\Omega_{\mathcal{B}}|.$$

The set $\Omega_{\mathcal{B}}$ may be either:

(a) the direct list of candidate pairs returned by `blocking`; or

(b) the union of within-block Cartesian products if the workflow first assigns records to blocks and then expands records inside each block.

All distributions, estimated counts, scores, and constraints in this document are defined on $\Omega_{\mathcal{B}}$. Therefore, the method is conditional on the pair having survived blocking. It cannot recover a true match that was not included in $\Omega_{\mathcal{B}}$.

The candidate-pair representation should contain, at minimum, identifiers for the record from A, the record from B, and the comparison vector γ_{ab} for each candidate pair. If candidate pairs are produced in multiple ways, duplicate pairs should be removed or handled consistently before the MEC iteration so that the same pair is not counted multiple times unless this behavior is explicitly intended.

4 Rationale for the Inverted Density Ratio

The key methodological change is that the blocking candidate space may be much more enriched in true matches than the full Cartesian product. In the classical MEC setting, the background pair space is approximately representative of nonmatches. In the blocking-based setting considered here, the candidate-pair space may instead be approximately representative of matches.

Accordingly, the proposed algorithm reverses the density ratio at the estimation stage. Define

$$s(\gamma) = \frac{u(\gamma)}{m(\gamma)}.$$

Large values of $s(\gamma)$ indicate that the pair looks like a nonmatch. Small values of $s(\gamma)$ indicate that the pair looks like a match.

The classification step is not reversed. The algorithm estimates nonmatch probabilities using $s = u/m$, but it still constructs a one-to-one set of matches. Nonmatches do not satisfy a one-to-one structure: one record can be a nonmatch with many records in the opposite file. Therefore, the one-to-one constraint must be applied to the selected match set, not to the nonmatch set.

5 Supported Comparison Models

The existing MEC machinery should be reused as much as possible, with the roles of the match and nonmatch distributions reversed in the density-ratio construction.

5.1 Binary MEC Model

For binary comparisons, assume that

$$\gamma_{ab}^k = 1$$

means agreement on the kth key variable and

$$\gamma_{ab}^k = 0$$

means disagreement. Under conditional independence of comparison components,

$$m(\gamma; \theta) = \prod_{k=1}^{K} \theta_k^{\gamma^k} (1 - \theta_k)^{1-\gamma^k},$$

and

$$u(\gamma; \xi) = \prod_{k=1}^{K} \xi_k^{\gamma^k} (1 - \xi_k)^{1-\gamma^k},$$

where

$$\theta_k = P(\gamma^k = 1 \mid M), \qquad \xi_k = P(\gamma^k = 1 \mid U).$$

In the inverted blocking-based algorithm, estimate the match parameters from all candidate pairs:

$$\widehat{\theta}_k = \frac{1}{N} \sum_{(a,b) \in \Omega_{\mathcal{B}}} \gamma_{ab}^k.$$

At iteration t, estimate the nonmatch parameters from the current nonmatch set $U^{(t)}$:

$$\widehat{\xi}_k^{(t)} = \frac{1}{|U^{(t)}|} \sum_{(a,b) \in U^{(t)}} \gamma_{ab}^k.$$

Then compute

$$s^{(t)}(\gamma_{ab}) = \frac{u(\gamma_{ab}; \widehat{\xi}^{(t)})}{m(\gamma_{ab}; \widehat{\theta})}.$$

Any existing smoothing or boundary handling used by the current binary MEC implementation for zero or one empirical probabilities should be preserved.

5.2 Continuous Parametric MEC Model

For continuous comparisons, assume that comparison components are nonnegative dissimilarities. For example, a text comparison may be represented as one minus a string similarity, and a numeric comparison may be represented as a nonnegative distance. The continuous parametric model uses a hurdle Gamma density for each component:

$$f(x; p_0, \alpha, \beta) = p_0^{I(x=0)} \left[(1 - p_0) v(x; \alpha, \beta)\right]^{I(x>0)}, \qquad x \geq 0,$$

where

$$v(x; \alpha, \beta) = \frac{\beta^{\alpha} x^{\alpha-1} \exp(-\beta x)}{\Gamma(\alpha)}.$$

Under conditional independence of comparison components,

$$m(\gamma; \theta) = \prod_{k=1}^{K} f(\gamma^k; p_{0M}^k, \alpha_M^k, \beta_M^k),$$

and

$$u(\gamma; \xi) = \prod_{k=1}^{K} f(\gamma^k; p_{0U}^k, \alpha_U^k, \beta_U^k).$$

In the inverted blocking-based algorithm, estimate the match-side hurdle Gamma parameters from all candidate pairs in $\Omega_{\mathcal{B}}$. For each component k,

$$\widehat{p}_{0M}^k = \frac{1}{N} \sum_{(a,b) \in \Omega_{\mathcal{B}}} I(\gamma_{ab}^k = 0),$$

and estimate (α_M^k, β_M^k) from the positive values of γ^k over $\Omega_{\mathcal{B}}$ using the same Gamma maximum-likelihood procedure as in the existing continuous parametric MEC implementation.

At iteration t, estimate the nonmatch-side hurdle Gamma parameters from the current nonmatch set $U^{(t)}$. For each component k,

$$\widehat{p}_{0U}^{k(t)} = \frac{1}{|U^{(t)}|} \sum_{(a,b) \in U^{(t)}} I(\gamma_{ab}^k = 0),$$

and estimate $(\alpha_U^{k(t)}, \beta_U^{k(t)})$ from the positive values of γ^k inside $U^{(t)}$.

The inverted ratio is then

$$s^{(t)}(\gamma_{ab}) = \frac{u(\gamma_{ab}; \widehat{\xi}^{(t)})}{m(\gamma_{ab}; \widehat{\theta})}.$$

For mixed binary and continuous parametric comparisons, the overall density ratio should be formed as the product of the corresponding binary and continuous component contributions, using the same conditional-independence convention as the existing MEC implementation.

6 Initialization by Disagreement Norms

The algorithm requires an initial matched set $M^{(0)}$ and an initial nonmatch set $U^{(0)}$. The initial nonmatch set must contain all candidate pairs that lose one-to-one conflicts against better candidate pairs. The proposed initialization uses only a disagreement norm.

For initialization only, construct a disagreement-oriented vector

$$\widetilde{\gamma}_{ab}.$$

For continuous dissimilarity components already represented as distances, set

$$\widetilde{\gamma}_{ab}^{k} = \gamma_{ab}^{k}.$$

For binary agreement components, invert the coding when computing the norm:

$$\widetilde{\gamma}_{ab}^{k} = 1 - \gamma_{ab}^{k}.$$

Thus, for the initialization norm, zero means no disagreement and one means binary disagreement.

Compute the unweighted Euclidean norm

$$d_{ab} = \|\widetilde{\gamma}_{ab}\|_2 = \left(\sum_{k=1}^{K} (\widetilde{\gamma}_{ab}^{k})^2 \right)^{1/2}.$$

No additional component weights are introduced.

The initial matched set is obtained greedily:

1. Sort all pairs in $\Omega_{\mathcal{B}}$ in ascending order of d_{ab}.
2. Traverse the sorted list.
3. Add pair (a, b) to $M^{(0)}$ if neither a nor b has already been used in a previously selected pair.
4. Otherwise, skip the pair.

Then define

$$U^{(0)} = \Omega_{\mathcal{B}} \setminus M^{(0)}.$$

If blocks are disjoint and records cannot occur across multiple blocks, the greedy initialization may be applied blockwise and then combined. If a record can occur in several blocks or several candidate-pair sources, the one-to-one constraint must be enforced globally on $\Omega_{\mathcal{B}}$.

Pairs placed in $U^{(0)}$ are not permanently excluded. They are used only to initialize the nonmatch distribution. In later iterations, any pair in $\Omega_{\mathcal{B}}$ may enter the matched set if it is selected by the inverted MEC ranking and does not violate the one-to-one constraint.

7 Inverted MEC Iteration

The match distribution $\widehat{m}$ is estimated once from all candidate pairs in $\Omega_{\mathcal{B}}$. The nonmatch distribution $\widehat{u}^{(t)}$ is estimated iteratively from $U^{(t)}$. At each iteration, compute

$$s^{(t)}(\gamma_{ab}) = \frac{\widehat{u}^{(t)}(\gamma_{ab})}{\widehat{m}(\gamma_{ab})}.$$

Let

$$q_{ab}^{(t)}$$

denote the estimated probability that candidate pair (a, b) is a nonmatch. The inverted MEC analogue of the classical posterior formula is

$$\widehat{q}_{ab}^{(t)} = \min\left\{ \frac{|U^{(t)}|s^{(t)}(\gamma_{ab})}{|U^{(t)}|\left(s^{(t)}(\gamma_{ab}) - 1\right) + N}, 1 \right\}.$$

Then estimate the number of nonmatches by

$$\widetilde{n}_U^{(t)} = \sum_{(a,b)\in\Omega_{\mathcal{B}}} \widehat{q}_{ab}^{(t)}.$$

The corresponding estimated match probability is

$$\widehat{p}_{ab}^{(t)} = 1 - \widehat{q}_{ab}^{(t)}.$$

Numerical protections from the current MEC implementation should be reused. In particular, density values should be handled safely when an empirical component probability or a fitted density is zero or numerically negligible.

8 One-to-One Lower Bound on Nonmatches

The one-to-one constraint implies that not all candidate pairs in $\Omega_{\mathcal{B}}$ can be matches. Let

$$\nu$$

denote the maximum possible number of pairs that can be selected from $\Omega_{\mathcal{B}}$ while satisfying the one-to-one constraint. Then the number of nonmatches must be at least

$$n_U^{\min} = N - \nu.$$

If $\Omega_{\mathcal{B}}$ is generated as a union of full Cartesian products inside disjoint blocks, then

$$\nu = \sum_c \min(|A_c|, |B_c|),$$

where A_c and B_c are the records from files A and B inside block c.

If $\Omega_{\mathcal{B}}$ is an arbitrary candidate-pair graph, then ν is the cardinality of a maximum feasible one-to-one matching in that graph.

After computing $\widetilde{n}_U^{(t)}$, convert it to an integer count using the same count rounding convention as the existing MEC implementation. If no convention is already present, use nearest-integer rounding. Then apply the structural correction

$$\widehat{n}_U^{(t)} = \max\left\{ n_U^{\min},\ \min\left\{ N, \mathrm{round}(\widetilde{n}_U^{(t)}) \right\} \right\}.$$

The estimated number of matches is then

$$\widehat{n}_M^{(t)} = N - \widehat{n}_U^{(t)}.$$

9 Construction of the Updated Match Set

Although the algorithm estimates nonmatch probabilities, the classification set constructed at the end of each iteration is a set of matches. Since

$$s^{(t)}(\gamma) = \frac{u^{(t)}(\gamma)}{m(\gamma)},$$

pairs with smaller $s^{(t)}$ are more match-like. Therefore, the updated matched set $M^{(t+1)}$ is obtained as follows:

1. Sort all pairs in $\Omega_{\mathcal{B}}$ in ascending order of $s^{(t)}(\gamma_{ab})$.
2. Traverse the sorted list.
3. Add pair (a, b) to $M^{(t+1)}$ if neither a nor b has already been used in a previously selected pair.
4. Stop when $\widehat{n}_M^{(t)}$ pairs have been selected, or when no more feasible pairs are available.

Then set

$$U^{(t+1)} = \Omega_{\mathcal{B}} \setminus M^{(t+1)}.$$

This is the direct analogue of the classical MEC set construction. Classical MEC sorts in descending order of $r = m/u$. The proposed algorithm sorts in ascending order of $s = u/m$. These are equivalent rankings of match-likeness, but the latter is consistent with estimating nonmatch probabilities in the inverted procedure.

10 Complete Algorithm

Step 1. Construct the candidate-pair space. Receive or construct $\Omega_{\mathcal{B}}$ from the blocking stage. Set $N = |\Omega_{\mathcal{B}}|$.

Step 2. Compute comparison vectors. For every $(a, b) \in \Omega_{\mathcal{B}}$, compute the model comparison vector γ_{ab}.

Step 3. Compute initialization norms. Construct $\widetilde{\gamma}_{ab}$ for the norm calculation. Continuous dissimilarities are left unchanged. Binary agreement indicators are inverted. Compute $d_{ab} = \|\widetilde{\gamma}_{ab}\|_2$.

Step 4. Initialize matches and nonmatches. Sort candidate pairs in ascending order of d_{ab}. Greedily select a one-to-one set $M^{(0)}$. Set $U^{(0)} = \Omega_{\mathcal{B}} \setminus M^{(0)}$.

Step 5. Estimate the match distribution. Estimate $\widehat{m}$ from all candidate pairs in $\Omega_{\mathcal{B}}$, using the binary model or the continuous parametric hurdle Gamma model as appropriate.

Step 6. Iterate. For $t = 0, 1, 2, \ldots$:

(a) Estimate $\widehat{u}^{(t)}$ from $U^{(t)}$.

(b) Compute $s^{(t)} = \widehat{u}^{(t)}/\widehat{m}$ for all candidate pairs.

(c) Compute $\widehat{q}_{ab}^{(t)}$ for all candidate pairs.

(d) Compute $\widetilde{n}_U^{(t)} = \sum_{(a,b)} \widehat{q}_{ab}^{(t)}$.

(e) Apply the one-to-one lower bound to obtain $\widehat{n}_U^{(t)}$.

(f) Set $\widehat{n}_M^{(t)} = N - \widehat{n}_U^{(t)}$.

(g) Construct $M^{(t+1)}$ by greedy one-to-one deduplication in ascending order of $s^{(t)}(\gamma_{ab})$.

(h) Set $U^{(t+1)} = \Omega_{\mathcal{B}} \setminus M^{(t+1)}$.

Step 7. Stop. Stop when the estimated class counts stabilize, for example when

$$|\widehat{n}_U^{(t+1)} - \widehat{n}_U^{(t)}| < \delta,$$

or when the selected match set no longer changes. The convergence logic may mirror the existing MEC implementation.

11 Required Methodological Changes Relative to Current MEC

The implementation should follow these methodological changes:

1. Use the blocking candidate-pair space $\Omega_{\mathcal{B}}$ as the pair space for both training and prediction.
2. Replace the classical ratio $r = m/u$ by the inverted ratio $s = u/m$.
3. Estimate the match distribution m from all candidate pairs in $\Omega_{\mathcal{B}}$.
4. Estimate the nonmatch distribution $u^{(t)}$ from the current nonmatch set $U^{(t)}$.
5. Initialize $M^{(0)}$ by greedy one-to-one selection in ascending order of the disagreement norm.
6. Invert binary agreement indicators only for the initialization norm calculation.
7. Interpret the MEC posterior formula as estimating nonmatch probabilities.
8. Enforce the lower bound $\widehat{n}_U \geq N - \nu$.
9. Convert the estimated nonmatch count into an estimated match count via $\widehat{n}_M = N - \widehat{n}_U$.
10. Construct the updated match set by greedy one-to-one deduplication in ascending order of $s = u/m$.
11. Define the updated nonmatch set as the complement of the selected match set.

12 Methodological Invariants for Validation

The following invariants should hold for every completed iteration:

(i) $M^{(t)}$ satisfies the one-to-one constraint.

(ii) $U^{(t)} = \Omega_{\mathcal{B}} \setminus M^{(t)}$.

(iii) $\widehat{n}_U^{(t)} \geq N - \nu$.

(iv) $\widehat{n}_M^{(t)} = N - \widehat{n}_U^{(t)}$.

(v) The updated match set is selected in ascending order of $s = u/m$.

(vi) The binary inversion $\widetilde{\gamma}^k = 1 - \gamma^k$ is used for initialization norms when γ^k is a binary agreement indicator.

(vii) Nonmatches are not deduplicated. Deduplication is applied only when constructing a candidate match set.

(viii) Pairs initially placed in $U^{(0)}$ can re-enter a later matched set if selected by the inverted MEC ranking.

(ix) No pair outside $\Omega_{\mathcal{B}}$ can be selected as a match.

13 Edge Cases and Methodological Warnings

If $U^{(0)}$ is empty or too small for the selected binary or continuous parametric model to estimate u, the procedure cannot be started reliably. The implementation should report an informative failure or use an explicitly documented fallback already present in the package. The method should not silently fabricate additional nonmatches.

If the candidate-pair space is not sufficiently enriched in matches, estimating m from all candidate pairs may be biased. This is a limitation of the inverted approach and should be reported through diagnostics when possible.

If blocks overlap or the same pair can be generated more than once, duplicate handling must be deterministic. The statistical interpretation of N, $\widehat{n}_U$, and $\widehat{n}_M$ requires a well-defined candidate-pair space.

Ties in initialization norms or inverted ratio scores should be resolved deterministically, for example by stable ordering of the pair identifiers. This is important for reproducibility.

Blocking-stage false negatives are outside the scope of the MEC iteration. If a true match is not included in $\Omega_{\mathcal{B}}$, the proposed algorithm cannot select it.

14 Summary

The proposed method is an inverted MEC procedure on a blocking candidate-pair space. The central idea is to treat all candidate pairs from blocking as the empirical basis for estimating the match distribution, while estimating the nonmatch distribution from the current complement of the one-to-one matched set. The inverted ratio

$$s = \frac{u}{m}$$

produces nonmatch probabilities and an estimated number of nonmatches. The algorithm then converts this into an estimated number of matches and constructs the updated match set by greedy one-to-one deduplication in ascending order of s.

The procedure remains close to classical MEC: it keeps the greedy MEC-set construction, the one-to-one classification set, and the posterior count-estimation logic. The only substantive changes are the blocking-based pair space, the reversed density ratio, the estimation of m from all candidate pairs, and the initialization based on disagreement norms.

15 Simulation Study

The method should be tested using the `census` and `census` datasets. The code should be similar to that included in the vignette.

16 Acceptance Criteria

1. Both FLR and MMR are below 2%.
2. `R CMD check` passes with 0 errors and 0 warnings.
3. The package installs succesfully.

References

[1] Beresewicz, M. and Struzik, A. (2026). *blocking: An R Package for Blocking of Records for Record Linkage and Deduplication*. Manuscript.

[2] Choi, S. C. and Wette, R. (1969). Maximum likelihood estimation of the parameters of the Gamma distribution and their bias. *Technometrics*, 11(4), 683–690.

[3] Fellegi, I. P. and Sunter, A. B. (1969). A theory for record linkage. *Journal of the American Statistical Association*, 64(328), 1183–1210.

[4] Lee, D., Zhang, L.-C., and Kim, J. K. (2022). Maximum entropy classification for record linkage. *Survey Methodology*, 48(1).

[5] Qin, T. and Xu, Y. (2026). StatsClaw: An AI-collaborative workflow for statistical software development. arXiv preprint.

[6] Struzik, A. and Beresewicz, M. (2026). Capturing small discrepancies in record linkage: A maximum entropy framework with continuous similarity measures. Manuscript.

[7] Vo, T. H., Chauvet, G., Happe, A., Oger, E., Paquelet, S., and Gares, V. (2023). Extending the Fellegi–Sunter record linkage model for mixed-type data with application to the French national health data system. *Computational Statistics & Data Analysis*, 179.